mini-scope – User Guide

Watch and record dsPIC33CK variables live, without halting the CPU

October 22, 2026

1. What is mini-scope for?

A debugger has to halt the CPU to read variables, which is harmful in a motor control application. mini-scope reads the variables while the application keeps running: a small module in the controller samples the selected variables at a fixed rate and sends them over a UART to the PC, where they are shown as curves in the browser.

There are two modes:

- **Stream:** continuous display, like a chart recorder. Suited to slow signals such as speed, temperature or states.
- **Capture:** records a short time window at any sample rate, started by a trigger such as a current rise or a fault. Works like a storage oscilloscope.

2. What you need

Item	Purpose
Python 3.9 or later	PC user interface; the first start installs everything else
USB-UART adapter, 3.3 V, 115200 baud	connection to the board, galvanically isolated if possible
`scope.c` and `scope.h` in the firmware project	module in the controller
the project's ELF file	lets you pick variables by name

It also works without hardware, see section 7.

3. Preparing the firmware

1. Add `firmware\scope.c` and `firmware\scope.h` to the MPLAB X project.
2. Call `scope_init();` once in `main()`, after the clock and pin setup. `scope_init()` configures PPS, UART, timer and interrupts. Nothing else is needed.
3. If your application already uses UART1 or Timer1, select a free one at the top of `scope.h`: `SCOPE_UART` 1, 2 or 3, and `SCOPE_TIMER` `SCOPE_T1`, `SCOPE_CCP1` … `SCOPE_CCP4` or `SCOPE_NONE`. To use other pins than the ICSP header, set `SCOPE_TX_RP` and `SCOPE_RX_RP` (RB*n* = RP(32 + *n*), RC*n* = RP(48 + *n*)). Direction and digital mode are set automatically. The values are listed in section 10.
4. If RAM is tight, reduce the buffers in `scope.h`. By default mini-scope uses about 4.7 KB. With `SCOPE_CAPBUF 1024u`, `SCOPE_TXBUF 256u` and `SCOPE_MAX_CH 8` it uses about 1.4 KB. The full table is in `README.md`. The user interface adapts automatically.
5. Check the peripheral clock: `SCOPE_FP_HZ` in `scope.h` must equal FOSC/2. The default is 32 MHz.
6. Build and program.

All settings are described in `README.md`.

4. Connecting

By default the data runs over the **ICSP header**: TX on PGD (RB5), RX on PGC (RB6).

1. Program the firmware with the PICkit, then **disconnect the PICkit**.

2. Connect the USB-UART adapter to the ICSP header:

ICSP pin	Signal	Adapter
3	GND	GND
4	PGD = TX from the controller	RX
5	PGC = RX to the controller	TX
2	3V3	**do not** connect

3. Power the board from its own supply.

Debugging and mini-scope cannot run at the same time, because both use the same pins.

5. Starting the user interface

Double-click `start_gui.bat`.

On the first start it looks for Python 3.9 or later, creates a Python environment `.venv` in the project folder and installs the required packages. This takes about a minute and needs internet access (behind a proxy, set `HTTPS_PROXY` first). Later starts take a few seconds. The browser opens at http://localhost:8080.

From top to bottom the page has these areas:

Area	Contents
Connection	Port, Baud, status line, **Connect** button
ELF file	path to the ELF, **Load** button
Channels	one row per variable: Variable, Address, type, Factor, delete
Settings	Stream or Capture, Sample period, trigger
Buttons	**Start**, **Stop**, **CSV**
Chart	curves, legend, zoom slider

6. Step by step

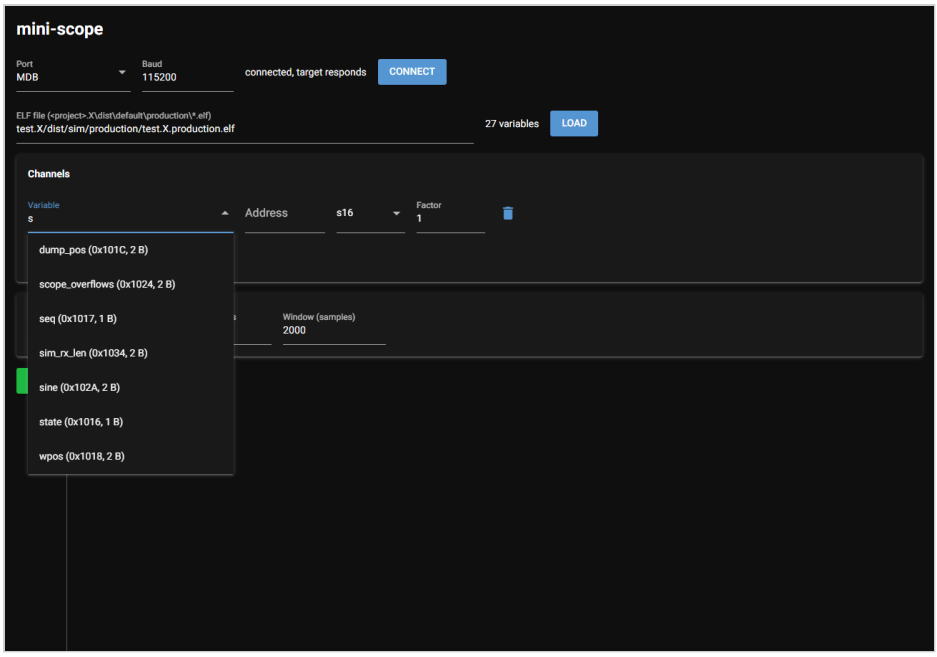
6.1 Connect

1. Under **Port**, select the adapter's COM port.
2. Leave **Baud** at 115200 unless the firmware uses a different rate.
3. Click **Connect**. The status line must show **"connected, target responds"**. The button now reads **Disconnect**, and Port and Baud are locked while connected.
4. **Disconnect** stops a running measurement and releases the COM port, for example to program the board again.

6.2 Select variables

1. Click **Browse ...** and select the ELF file of your project in the file dialog. It is usually `...\MyProject.X\dist\default\production\MyProject.X.production.elf`. The file is loaded right away. Alternatively, type or paste the path into **ELF file** and click **Load**. The number of variables found appears next to the buttons.
2. Click the **Variable** field. A list of all variables with their address and size drops down. Typing filters the list.

3. Click a variable. **Address** and type are filled in automatically.
4. Add more variables with **+ Channel**, remove them with the trash can.



Selecting variables after loading the ELF

The **type** defines how the bytes are read:

Type	Meaning
`s8`, `s16`, `s32`	signed, for example `int16_t`
`u8`, `u16`, `u32`	unsigned, for example `uint16_t`
`f32`	`float`

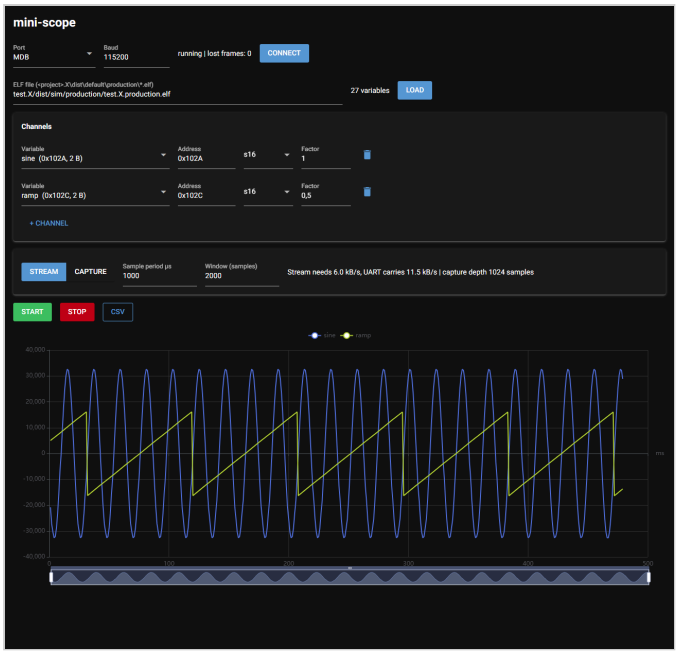
A signed type is suggested. For `unsigned` variables, change it to `u…` by hand.

Factor multiplies the value for display. Use it to convert to physical units, for example Q15 to amperes, or to bring signals of different magnitude onto one scale. The factor only affects the display and the CSV export, not the trigger.

Without an ELF file, enter the **Address** directly, for example from the `.map` file, and choose the type by hand.

6.3 Watch values continuously (Stream)

1. Select **Stream**.
2. Enter the **Sample period µs**, for example 1000 for 1 kHz.
3. Set **Window (samples)** to the number of samples to show.
4. Click **Start**.



Stream view

To the right of the settings, mini-scope shows the data rate the stream needs and the rate the UART can carry. If the sample period is too short, an orange warning appears instead: **"⚠ Too fast for the UART …"**, together with the shortest sample period that still fits. Clicking **Start** anyway shows the warning again; samples are then dropped and the status line counts the **lost frames**. Increase the period, use fewer channels, or use Capture instead.

6.4 Record an event (Capture)

1. Select **Capture**.
2. Enter the **Sample period µs**. A few µs are possible here.
3. Set up the trigger:
    - **Trigger channel:** number of the channel in the list, the first one is 0.
    - **Edge:** rising, falling or none. "None" records immediately.
    - **Threshold (raw):** a **raw value**, without the factor.
    - **Pretrigger %:** how much of the recording lies before the trigger.
    - **Auto re-arm:** when checked, the trigger is re-armed after every recording. Unchecked, you get exactly one recording.
4. Click **Start**. As soon as the condition occurs, the recording appears. The trigger point is at **0 ms**.



Capture on a rising edge through 0, 20 % pretrigger

The recording length is the capture buffer (4 KB by default) divided by the bytes per sample. It is shown to the right of the settings as **capture depth**. Fewer channels mean a longer recording.

6.5 Working with the chart

- **Zoom:** mouse wheel in the chart, or drag the slider below it.
- **Hide a channel:** click its name in the legend.
- **Read values:** hover over the curve.

6.6 Saving data

CSV saves the data currently shown, that is the stream window or the last recording. The first column is the time in ms; the file is comma separated with a decimal point.

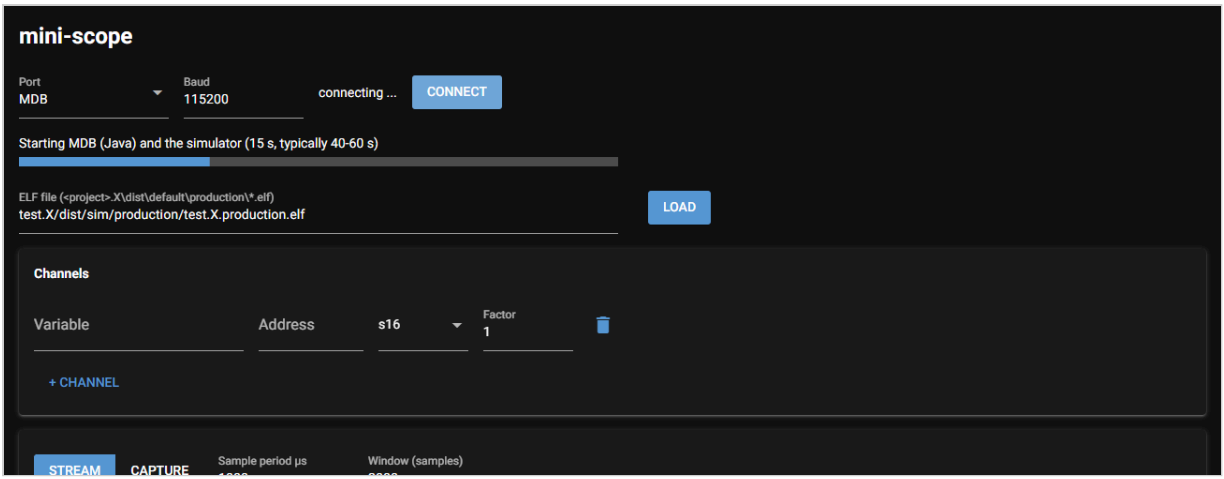
6.7 Quitting

Click **Stop**, then close the console window or press Ctrl+C.

7. Simulation without hardware

Double-click `start_sim.bat`. It builds the test project `test.X` for the simulator and starts the user interface.

1. Port **MDB** is preselected. Click **Connect**.
2. Wait until the progress bar has finished, typically 40–60 s. During this time MPLAB X starts the simulator in the background and loads the firmware.
3. Click **Load**. The ELF path is already filled in.
4. Select variables, for example `sine` and `ramp`, then continue as in section 6.



Progress bar while the simulator starts

The simulator runs much slower than real time, and data appears with a delay of a few seconds. That is good enough for a functional test. Details are in `SIMULATOR.md`.

8. Troubleshooting

The most common problems are listed below. `TROUBLESHOOTING.md` goes through all of them step by step, including how to find out the real clock frequency.

Symptom	Cause and remedy
"Python 3.9 or newer was not found"	install Python from python.org with "Add python.exe to PATH" ticked
"Package installation failed"	no internet access, or a proxy: set `HTTPS_PROXY=http://proxy:port` and start again
"port open, but no answer from target"	RX and TX swapped, wrong baud rate, `scope_init()` not called, or `SCOPE_FP_HZ` does not match the clock
Port missing from the list	plug in the adapter and reload the page (F5)
"Target rejected the configuration"	too many channels (`SCOPE_MAX_CH`, default 16) or an odd address on a 16- or 32-bit channel
Curves jump, lost frames increase	stream too fast: increase the period or use Capture
No capture appears	the trigger condition never occurs: check the threshold as a raw value, or try edge "none"
Values look wrong	wrong type, for example `s16` instead of `u16`, or `s32` instead of `f32`
Variables no longer match after rebuilding the firmware	**Load** the ELF again, the addresses may have moved
Linker error `multiple definition of _T1Interrupt` (or `_U1RXInterrupt` ...)	the application already uses this timer or UART: select another one in `scope.h`
Simulator: progress stalls for more than 90 s	close the console window and run `start_sim.bat` again

9. Good to know

- 32-bit values are read in two steps. If a higher-priority interrupt writes the variable in between, a single value can be wrong.
- If your code updates a variable in several steps (first set, then correct), mini-scope may catch the intermediate value. Compute the value in a local variable first and write it with a single assignment.

- With very short sample periods, every sample costs an interrupt. In a motor control application, set `SCOPE_TIMER = SCOPE_NONE` and call `scope_sample()` at the end of the control ISR. Sampling is then synchronous to the PWM.

10. Appendix: UART pins

Values for `SCOPE_TX_RP` and `SCOPE_RX_RP` in `scope.h`, for the dsPIC33CK64MC103 (36 pins). The default is RB5/RB6 on the ICSP header.

Port pin	`SCOPE_TX_RP` / `SCOPE_RX_RP`	Package pin (36-pin)	Other functions on the pin	Note
RB0	32	17	OSCI, CLKI, AN5	not with a crystal or external clock
RB1	33	18	OSCO, CLKO, AN6	not with a crystal
RB2	34	19	AN1, AN7, ANA0, CMP1D, OA2OUT, INT0	analog pin, set to digital automatically
RB3	35	20	PGD2, AN8, OA2IN-	ICSP channel 2
RB4	36	21	PGC2, OA2IN+	ICSP channel 2
RB5	**37**	**24**	**PGD3**	**default TX: ICSP channel 3**
RB6	**38**	**25**	**PGC3**	**default RX: ICSP channel 3**
RB7	39	26	TDO, AN2	JTAG must be off (`JTAGEN = OFF`)
RB8	40	27	PGD1, AN10, SCL1	ICSP channel 1, I²C1
RB9	41	28	PGC1, AN11, SDA1	ICSP channel 1, I²C1
RB10	42	33	TMS, PWM3H	PWM output, JTAG
RB11	43	34	TCK, PWM3L	PWM output, JTAG
RB12	44	35	TDI, PWM2H	PWM output, JTAG
RB13	45	36	PWM2L	PWM output
RB14	46	1	PWM1H	PWM output
RB15	47	2	PWM1L	PWM output
RC0	48	4	AN12	analog pin, set to digital automatically
RC1	49	12	AN13, CMP1B, OA3IN-, ISRC0	analog pin, set to digital automatically
RC2	50	13	AN14, OA3IN+, ISRC1	analog pin, set to digital automatically
RC3	51	16	AN15, IBIAS2	analog pin, set to digital automatically
RC4	52	29	–	free digital pin
RC5	53	30	–	free digital pin

Port A (RA0 … RA4) is not remappable and cannot be used. Source: the device description in the dsPIC33CK-MC_DFP pack (`DSPIC33CK64MC103.PIC`); the package pin numbers apply to the 36-pin dsPIC33CK64MC103.

Notes:

- If the debugger uses the same ICSP channel (`ICS` configuration bit), do not debug while mini-scope runs on these pins.
- On motor control boards the PWM pins are usually taken by the power stage.